

ANNOVA ONE-WAY

PROBLEM:

Investigate whether there is a significant difference in the displacement (disp) of cars based on the number of forward gears (gear) they have. Perform a one-way ANOVA and interpret the results.

AIM:

To determine if there are significant differences in car displacement among cars with different numbers of cylinders in the `mtcars`.

CODE IN R LANGUAGE:

```
# Loading the package
library(dplyr)

#Perform oneway ANNOVA
head(mtcars)

mtcars_aov <- aov(mtcars$disp~factor(mtcars$gear))
summary(mtcars_aov)
```

OUTPUT:

```
head(mtcars)

##           mpg  cyl  disp  hp  drat   wt  qsec  vs  am  gear carb
## Mazda RX4      21.0   6  160  110  3.90  2.620 16.46  0   1    4    4
## Mazda RX4Wag   21.0   6  160  110  3.90  2.875 17.02  0   1    4    4
## Datsun 710      22.8   4  108   93  3.85  2.320 18.61  1   1    4    1
## Hornet 4 Drive  21.4   6  258  110  3.08  3.215 19.44  1   0    3    1
## Hornet Sportabout 18.7   8  360  175  3.15  3.440 17.02  0   0    3    2
## Valiant        18.1   6   225  105  2.76  3.460 20.22  1   0    3    1
```

```
Summary(mtcars_aov)
```

```
##              Df Sum Sq MeanSq Fvalue Pr(>F)
## factor(mtcars$gear) 2  280221  140110   20.73  2.56e-06 ***
## Residuals          29  195964    6757
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

RESULT:

The one-way ANOVA results indicate that there are significant differences in car displacement across different numbers of cylinders in the `mtcars` dataset, with cars having more cylinders generally exhibiting larger displacements.

ANNOVA TWO-WAY

PROBLEM:

Explore whether there is a significant interaction effect between the number of forward gears (gear) and the type of transmission (am) on the displacement (disp) of cars in the `mtcars` dataset. Perform a two-way ANOVA and interpret the interaction results.

AIM:

To investigate the interaction effect between the number of forward gears (gear) and the type of transmission (am) on car displacement (disp) in the `mtcars` dataset, aiming to understand how the combination of these two factors influences vehicle specifications.

CODE IN R LANGUAGE:

```
#Perform Two-way ANNOVA
#Calculate the test statistics using Aov function
mtcars_aov2 <- aov(mtcars$disp~factor(mtcars$gear) *
                  factor(mtcars$am))
summary(mtcars_aov2)
```

OUTPUT:

```
Summary(mtcars_aov2)

##              Df Sum Sq Mean Sq F value    Pr(>F)
## factor(mtcars$gear)  2  280221   140110   20.695 3.03e-06 ***
## factor(mtcars$am)    1    6399     6399    0.945  0.339
## Residuals          28  189565     6770
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

RESULT:

The two-way ANOVA results indicate a significant interaction effect between the number of gears and the type of transmission on car displacement in the `mtcars` dataset, suggesting that the relationship between these factors is not consistent across all combinations.